Distinguishable Dyads: Model Specification and Output

Pascal Küng

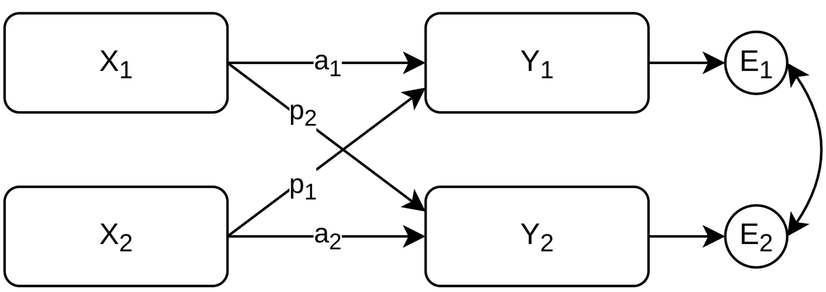
Full GitHub repository: The full repository contains the tutorial source files, reusable helper functions, model code, rendered PDFs, and additional information: https://github.com/Pascal-Kueng/05DyadicDataAnalysis

1 Background

Distinguishable dyads have partner roles that are meaningful for the model. In this tutorial example, the partners are distinguished by gender and represented with `is_male` and `is_female`.

For intensive longitudinal data, a distinguishable APIM needs to model:

- role-specific fixed effects, because the two partners have meaningful labels;
- role-specific average time trends and actor/partner effects;
- stable between-dyad differences in each role's average level and slopes;
- correlations among the role-specific random effects;
- role-specific residual variances, when the two roles may differ in unexplained variability;
- same-day residual interdependence between partners after fixed and random effects have been accounted for.



2 Data structure

Table 1: Example person-day structure for a distinguishable dyad

coupleID	userID	diaryday	couple_day	gender	member	is_female	is_male
31	31_1	0	31.0	female	1	1	0

coupleID	userID	diaryday	couple_day	gender	member	is_female	is_male
31	31_1	1	31.1	female	1	1	0
31	31_1	2	31.2	female	1	1	0
31	31_2	0	31.0	male	2	0	1
31	31_2	1	31.1	male	2	0	1
31	31_2	2	31.2	male	2	0	1

The dummy variables define partner role. They are used in the fixed effects, random effects, residual variance model, and residual covariance model.

3 glmmTMB model

```
model_dist_apim_long_glmmtmb <- glmmTMB(
  closeness ~ 0 + is_male + is_female +

    # Role-specific time slopes.
    is_male:diaryday_c +
    is_female:diaryday_c +

    # Between-person APIM effects.
    is_male:provided_support_actor_cbp +
    is_male:provided_support_partner_cbp +
    is_female:provided_support_actor_cbp +
    is_female:provided_support_partner_cbp +

    # Within-person APIM effects.
    is_male:provided_support_actor_cwp +
    is_male:provided_support_partner_cwp +
    is_female:provided_support_actor_cwp +
    is_female:provided_support_partner_cwp +

    # Dyad-level random intercepts and time slopes.
    (0 + is_male + is_female +
       is_male:diaryday_c +
       is_female:diaryday_c | coupleID) +

    # Same-day residual covariance between partners.
    # us() estimates partner-specific residual variances and their covariance.
    us(0 + is_male + is_female | coupleID:diaryday),

  # The Gaussian residual variance is represented by the us() block above.
  # Without dispformula = ~ 0, glmmTMB would add a second independent
  # residual variance on top of the dyadic residual covariance structure.
  # For families without a residual, dispformula should NOT be set to zero.
  dispformula = ~ 0,
  family = gaussian(),
  data = df_long_apim
)
```

4 brms model

```
formula_dist_apim_long_simple <- bf(
  closeness ~ 0 + is_male + is_female +

    # Role-specific time slopes.
    is_male:diaryday_c +
    is_female:diaryday_c +
```

```r
    # Between-person APIM effects.
    is_male:provided_support_actor_cbp +
    is_male:provided_support_partner_cbp +
    is_female:provided_support_actor_cbp +
    is_female:provided_support_partner_cbp +

    # Within-person APIM effects.
    is_male:provided_support_actor_cwp +
    is_male:provided_support_partner_cwp +
    is_female:provided_support_actor_cwp +
    is_female:provided_support_partner_cwp +

    # Dyad-level random intercepts and time slopes.
    (0 + is_male + is_female +
       is_male:diaryday_c +
       is_female:diaryday_c | coupleID) +

    # Native same-day residual correlation between partners.
    unstr(time = member, gr = couple_day),

  # Role-specific residual SDs.
  # brms estimates sigma coefficients on the log scale, so b_sigma_* draws
  # must be exponentiated before reporting residual SDs.
  sigma ~ 0 + is_male + is_female
)

priors_dist_apim_long_simple <- c(
  prior(normal(4, 2), class = "b", coef = "is_male"),
  prior(normal(4, 2), class = "b", coef = "is_female"),
  prior(normal(0, 2), class = "b"),
  prior(exponential(1), class = "sd"),
  prior(lkj(2), class = "cor"),
  prior(normal(0, 0.7), class = "b", dpar = "sigma"),
  prior(lkj(2), class = "cortime")
)

model_dist_apim_long_simple <- brm(
  formula = formula_dist_apim_long_simple,
  data = df_long_apim,
  family = gaussian(link = identity),
  prior = priors_dist_apim_long_simple,
  chains = 4,
  cores = 4,
  iter = 2000,
  warmup = 1000,
  seed = 123,
  file = file.path("brms_cache", "example1_dist_apim_long_simpel")
)
```

5 Output helper

The helper function `summarize_distinguishable_apim_brms()` used below is defined in `00_R_Functions/ReportModels.R` on GitHub. It summarizes fixed effects, random-effect SDs/correlations, and residual quantities. For `brms`, sigma coefficients are stored on the log scale, so the helper exponentiates `b_sigma_*` draws before computing residual SDs, variances, and same-day residual covariance.

For `glmmTMB`, the same residual quantities are available from the fitted variance-covariance output, but interval estimation is less direct because there are no posterior draws. One would usually transform parametric bootstrap draws or asymptotic draws from the fitted parameter covariance matrix.

```r
dist_results <- summarize_distinguishable_apim_brms(
  model_dist_apim_long_simple,
  gr = "coupleID"
)
```

6 Random-effect SDs

```
kable(
  dist_results$random_sd_summary,
  digits = 3,
  caption = "Random-effect standard deviations with 95% credible intervals"
)
```

Table 2: Random-effect standard deviations with 95% credible intervals

parameter	estimate	est_error	q_low	q_high
is_female	0.913	0.067	0.796	1.056
is_female:diaryday_c	0.016	0.001	0.014	0.019
is_male	0.846	0.061	0.737	0.972
is_male:diaryday_c	0.017	0.002	0.014	0.020

7 Random-effect correlations

```
kable(
  dist_results$random_cor_summary,
  digits = 3,
  caption = "Random-effect correlations with credible intervals",
  booktabs = TRUE
) |>
  kable_styling(latex_options = "scale_down", font_size = 7)
```

Table 3: Random-effect correlations with credible intervals

parameter	estimate	est_error	q_low	q_high
is_female with is_female:diaryday_c	0.503	0.081	0.330	0.644
is_female with is_male:diaryday_c	0.025	0.108	-0.188	0.229
is_male with is_female	0.173	0.094	-0.024	0.348
is_male with is_female:diaryday_c	0.187	0.099	-0.013	0.376
is_male with is_male:diaryday_c	0.507	0.089	0.323	0.668
is_male:diaryday_c with is_female:diaryday_c	-0.015	0.117	-0.254	0.213

8 Residual structure

```
kable(
  dist_results$residual_summary,
  digits = 3,
  caption = "Residual SDs, variances, and same-day residual association"
)
```

Table 4: Residual SDs, variances, and same-day residual association

section	parameter	estimate	est_error	q_low	q_high
Residual structure	residual_variance_is_female	0.521	0.010	0.501	0.542

section	parameter	estimate	est_error	q_low	q_high
Residual structure	residual_variance_is_male	1.260	0.025	1.213	1.309
Residual structure	same_day_residual_correlation	0.041	0.013	0.016	0.067
Residual structure	same_day_residual_covariance	0.033	0.011	0.013	0.055
Residual structure	sigma_is_female	0.722	0.007	0.708	0.736
Residual structure	sigma_is_male	1.122	0.011	1.101	1.144

9 Compact reporting summary

Table 5: Compact distinguishable-dyad reporting summary

section	parameter	estimate	est_error	q_low	q_high
Fixed effects	is_male	4.486	0.088	4.319	4.660
Fixed effects	is_female	5.738	0.090	5.567	5.911
Fixed effects	is_male:diaryday_c	-0.003	0.002	-0.007	0.001
Fixed effects	is_female:diaryday_c	0.011	0.002	0.007	0.014
Fixed effects	is_male:provided_support_actor_cbp	1.129	0.103	0.924	1.331
Fixed effects	is_male:provided_support_partner_cbp	0.308	0.099	0.110	0.503
Fixed effects	is_female:provided_support_actor_cbp	1.381	0.110	1.159	1.597
Fixed effects	is_female:provided_support_partner_cbp	0.591	0.110	0.379	0.811
Fixed effects	is_male:provided_support_actor_cwp	0.185	0.021	0.144	0.226
Fixed effects	is_male:provided_support_partner_cwp	0.125	0.021	0.083	0.166
Fixed effects	is_female:provided_support_actor_cwp	0.419	0.013	0.394	0.445
Fixed effects	is_female:provided_support_partner_cwp	0.170	0.013	0.145	0.197
Random-effect SDs	is_female	0.913	0.067	0.796	1.056
Random-effect SDs	is_female:diaryday_c	0.016	0.001	0.014	0.019
Random-effect SDs	is_male	0.846	0.061	0.737	0.972
Random-effect SDs	is_male:diaryday_c	0.017	0.002	0.014	0.020
Random-effect correlations	is_female with is_female:diaryday_c	0.503	0.081	0.330	0.644
Random-effect correlations	is_female with is_male:diaryday_c	0.025	0.108	-0.188	0.229
Random-effect correlations	is_male with is_female	0.173	0.094	-0.024	0.348
Random-effect correlations	is_male with is_female:diaryday_c	0.187	0.099	-0.013	0.376
Random-effect correlations	is_male with is_male:diaryday_c	0.507	0.089	0.323	0.668
Random-effect correlations	is_male:diaryday_c with is_female:diaryday_c	-0.015	0.117	-0.254	0.213
Residual structure	residual_variance_is_female	0.521	0.010	0.501	0.542
Residual structure	residual_variance_is_male	1.260	0.025	1.213	1.309
Residual structure	same_day_residual_correlation	0.041	0.013	0.016	0.067
Residual structure	same_day_residual_covariance	0.033	0.011	0.013	0.055
Residual structure	sigma_is_female	0.722	0.007	0.708	0.736
Residual structure	sigma_is_male	1.122	0.011	1.101	1.144